14,700 years old cannibals
Scientists discover a cave filled with symbolic bone carvings and skull cups, hinting at early cannibalism. **http://dgit.in/14700bone**

Bacteria for solar energy
Chemically modified bacteria covered in tiny solar panels could be the key in creating renewable solar fuels. **http://dgit.in/bacteriaSun**

in funding the mission. For starters, the cost is predicted to be around USD 1 million. These expenses will include a low-cost launch platform, orbital transfer options and other craft-integrity specs.

IARC isn't going to work on this mission alone and is already collaborating with other institutes and organisations. A few collaborators have officially signed onto the project including Satyabhama University from Chennai, Bellatrix Aerospace from Bangalore, and VIT University from Vellore. The collaborators have their own distinct roles to play in the project. Satyabhama University will be taking care of preparing the microbes and conducting Venus simulation tests. Bellatrix Aerospace is a private company developing orbital launch vehicles and electric propulsion systems. They'll be responsible of taking the nano-spacecraft to the Venusian orbit, at a lower cost compared to ISRO according to IARC. Finally, VIT University will also conduct research into preparing the microbes and work on dispersal mechanisms to inject the microbes into Venus' atmosphere.

WHY VENUS?

Venus was primarily chosen because of its close resemblance to Earth. According to Pushkar Vaidya, Mission Head of Bijayan Venus, "Terraforming makes sense for Venus more than any other planetary body in our solar system". Both the planets have a similar mass and size, and the most important point is its thick atmosphere. The idea of terraforming Mars has been going on for quite some time but the biggest problem is the absence of a considerable atmosphere on the red planet. Venus has a rich atmosphere composed of about 96.5% of carbon dioxide which pushes the surface pressure to 90 times greater than Earth. We already know that Venus is crazy hot, with an average surface temperature of 467 degrees Celsius. There's no magnetic field present on the planet due to its slow rotation, hence it's highly radioactive without any defense from the harmful rays from solar eruptions. If that wasn't enough to make life unfavourable, sulphuric acid rain is a common occurrence. Humans, or most organisms on Earth for that matter, won't be able to survive this environment, but there are few tough microorganisms living among us who might.

THE MICROBE PAYLOAD

Selecting the right organisms for this mission is highly critical for the ultimate goal of the mission – initiating terraforming of the planet. In order to achieve that there are two major challenges. The first is to reduce the amount of carbon dioxide in the atmosphere while the second is to lower the surface temperature. Carbon dioxide is a greenhouse gas, so it traps all the heat, and since it dominates the atmosphere, it's responsible for the crazy temperatures capable of melting lead. Hence, if the amount of carbon dioxide goes down, the temperatures will gradually fall down. The basic criteria for the microbes will be its capability of overcoming these two challenges. However, the extreme conditions of the atmosphere need to be taken into picture and survivability needs to be ensured. This will be the next set of criteria for the microbes.

Surviving in these extreme conditions isn't possible for regular organisms on Earth. Hence, it's a job for extremophiles. These microorganisms can survive and even thrive in extreme living conditions like high and low temperatures, high radiation zones, and even highly acidic areas. Considering the harsh atmospheric conditions of Venus, several extremophiles seem to fit a few of those. The possible candidates or rather the "chosen ones" to terraform Venus include cyanobacteria species, acidophiles, thermophiles, and polyextremophiles. Cyanobacteria is capable of photosynthesis and with an abundant supply of carbon dioxide in the atmosphere, they will be able to convert it into oxygen if they manage to thrive. However, they will require magnesium and phosphorus, and other nutrients in order to support photosynthesis, which seems unlikely. Acidophiles will be able to endure the harmful sulphuric acid in the clouds while thermophiles will be able to take on high temperatures (but not even close to the average surface temperature of Venus). The polyextremophile chosen for now are tardigrades or water bears. You're probably aware of these alien-looking bears that go viral every year, popular for their extreme survival capabilities, normally fatal for many organisms. They are the toughest microbes in the team, riding with high expectations on their tiny heads. These aren't the final microbes that will be sent, and we can expect more extremophiles to be added later after the Announcement of Opportunity (AO) is released in the upcoming months. As a rule, only non-pathogenic microbes will be used for the mission to avoid any harmful effects in the future.

Delivering the microbes into the Venusian atmosphere is riddled with risks where mistakes could be fatal for the mission. Rather than dropping off a lander or probe into the atmosphere, the